

Privacy Preserving Federated Learning

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Motivation

- Current privacy preserving federated learning frameworks can offer strong privacy guarantees, but these lead to increase time complexity
- Not all implementations may need that level of security or can tolerate the resulting slower speeds
- Federated frameworks need finer tuned controls over privacy settings to fit individual implementation needs



01

OVERVIEW

Federated Learning/
Definitions

02

ATTACKS

Attacks to Federated
Learning/Examples

03

SECURITY LEVELS

Defenses and
Techniques
Proposed



01

Overview

Federated Learning/ Definitions

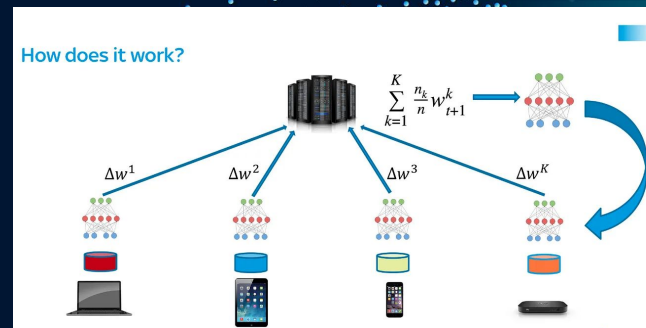
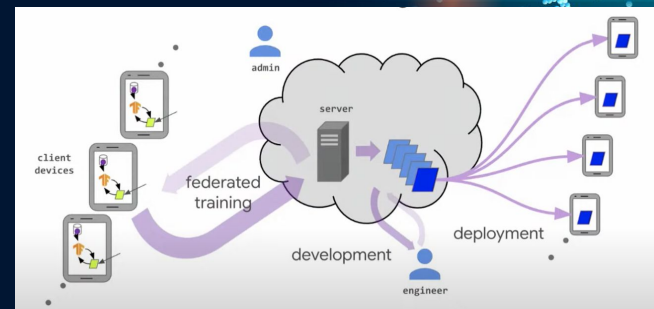
What is Federated Learning (FL)?

Coined by Google in 2016: “Federated Learning enables devices to collaboratively learn a shared prediction model while keeping all the training data on device, decoupling the ability to do machine learning from the need to store the data in the cloud”

FL considers data privacy in a decentralized collaborative learning setting.

Cross Device settings: Many clients and most clients participate once.

Cross silo settings: Few clients and most participate each round.

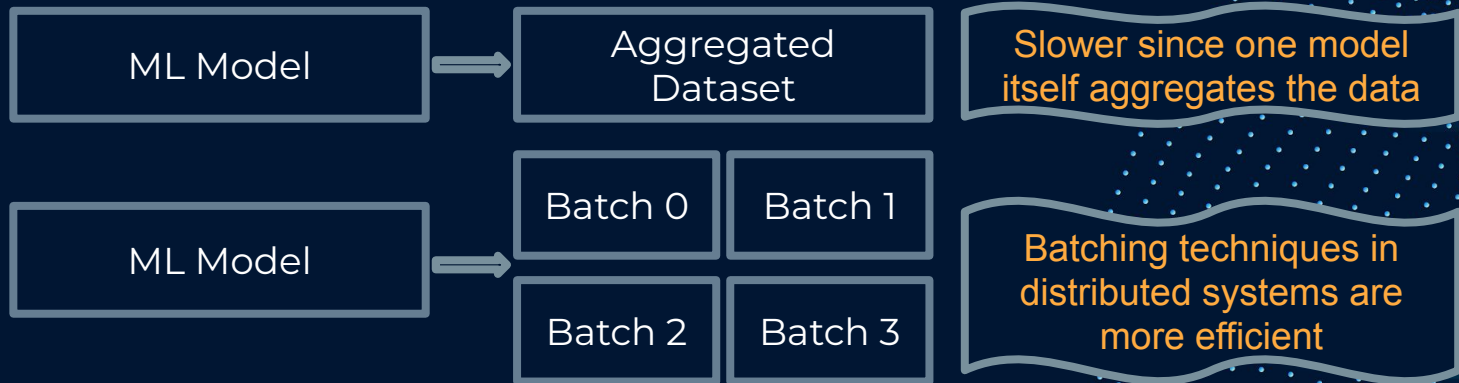


Benefits to FL

Security: Prevents sensitive user data from being stored in just one central server as part of training data

Real-Time Practices: Predictions happen on-device and may not even need internet connectivity

Time Efficient: Don't have to spend time aggregating all the data and then running an ML algorithm on the whole dataset



Privacy Techniques within FL

- Secure Multi-party Computation (SMPC): Parties only learn the final result, weights are encrypted.
- Differential Privacy: Adding noise
- Homomorphic Encryption

Combining methods further the privacy of the system.





02

Attacks

Attacks to Federated Learning/Examples

Possible Attacks to Privacy in FL

Compromised nodes (devices) sending malicious updates.

Devices training on compromised data (data poisoning)

Device drop out, data corruption during transmission

Inference time evasion attacks - leak local information during training

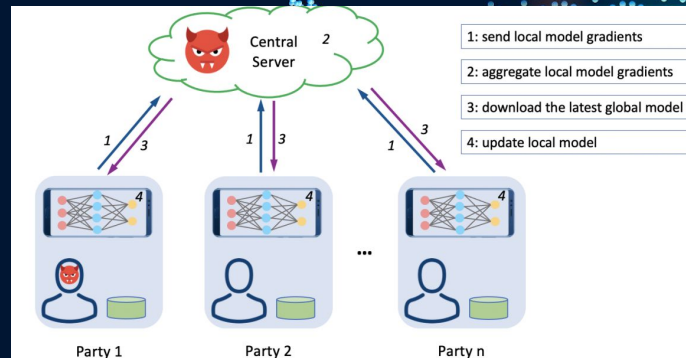


TABLE I: Classification of attacks in FL

	Training phase	Predicting phase
Local worker	- Data poisoning - Inference attack - Byzantine attack	- Evasion attack - Inference attack - Model extraction attack - Reconstruction attack
Center server	- Model poisoning - Reconstruction attack - Byzantine attack	
outsider	Inference attack	

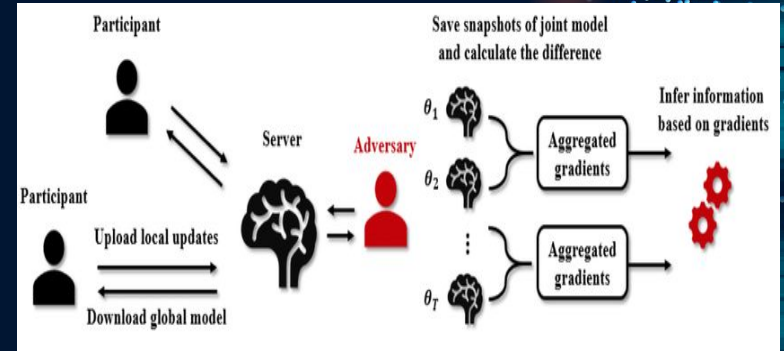
■ confidentiality ● integrity ▲ availability

Confidentiality

Inference Attack - Uses model to determine if sample belongs to training set, which could reveal information

Reconstruction Attack - Tries to obtain original training set information using model gradients, parameters, conf. values, etc.

Model Extraction Attack - Use information like confidence values, hyperparameters, similar data dist., etc. to reconstruct model

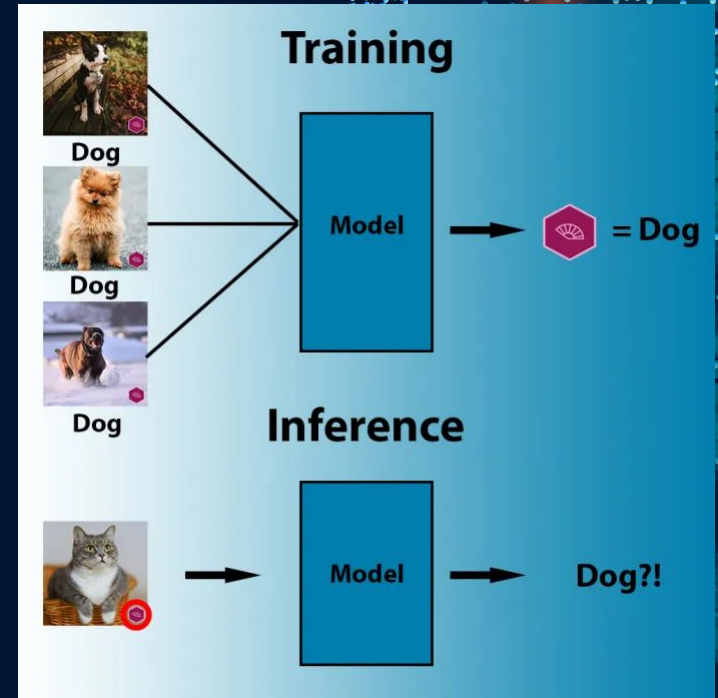


Integrity

Data Poisoning - Adding data to training set maliciously

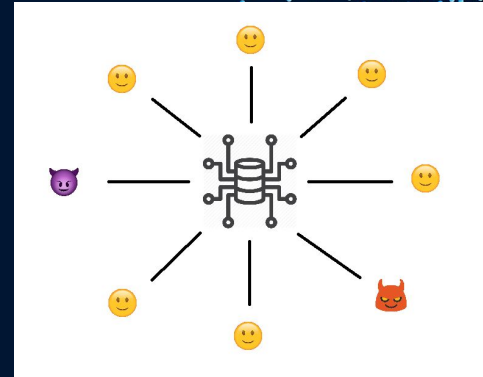
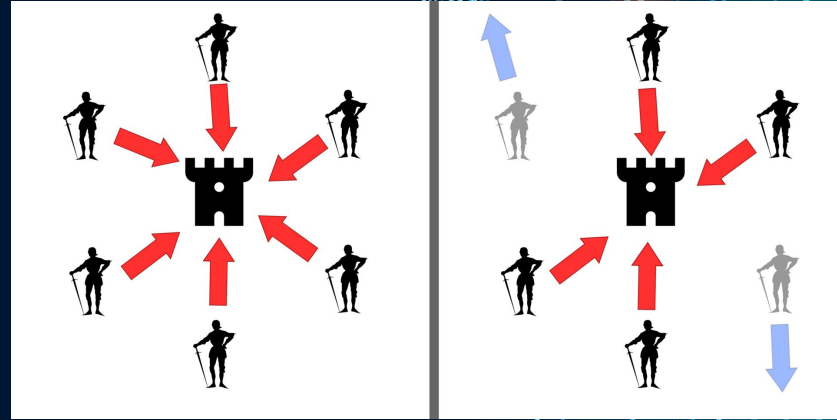
Model Poisoning - Changing parameters/weights of target model

Evasion Attack - Changing input data during prediction phase



Availability

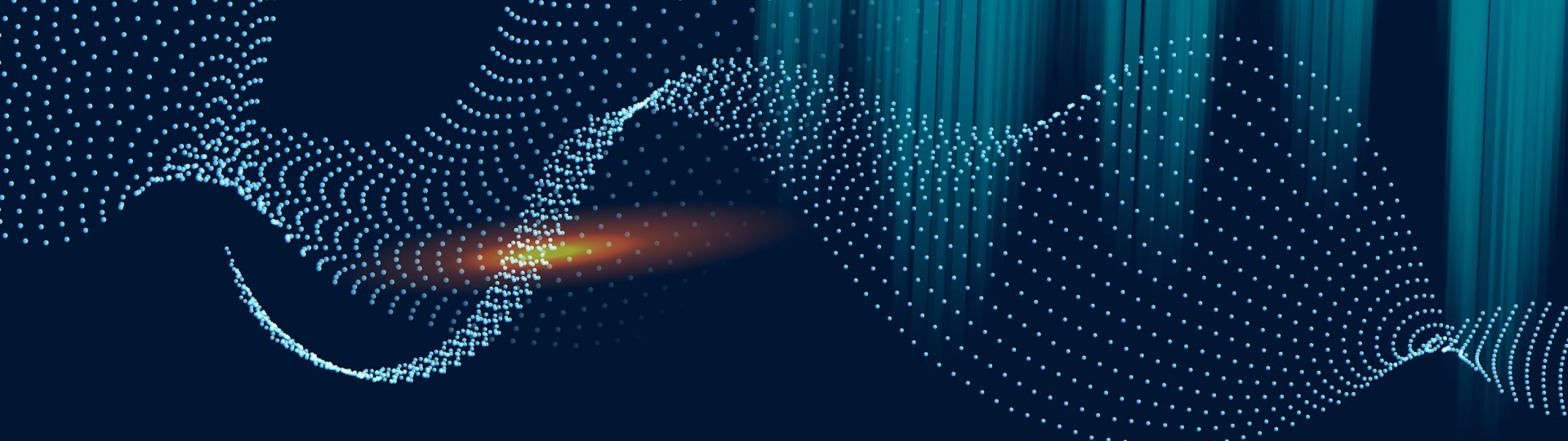
Byzantine Attack - Hides among FL members and arbitrarily uploads malicious data to destroy global model



Impact of Attacks

- **Confidentiality:** Personal information such as medical records, Personally-identifiable information (social security, location, etc.), and payment details may be accessed through these attacks -> costly, privacy violation, unsafe
- **Integrity:** Leads to poor model performance -> waste of time and resources; Can lead to a specific model fault, such as in autonomous vehicle systems or a medical model incorrectly predicting a disease
- **Availability:** Unreliability of nodes in the system that may go undetected





03

Security Levels

Defenses and Techniques Proposed

Security Definitions

Least Secure

Public

Fastest

Restricted

Sensitive

Most Secure

Secure

Slowest



Security Level Assurances

Security Level	Integrity	Encryption	Malicious Clients	Malicious Server
Public	✓	✗	✗	✗
Restricted	✓	✓	✗	✗
Sensitive	✓	✓	✓	✗
Secure	✓	✓	✓	✓

Defenses

Concern	Defense
Unauthorized Viewing	Encryption
Unauthorized Modification	Integrity Constraints
Untrusted Clients	Robust Aggregation
Untrusted Server	Differential Privacy Homomorphic Encryption

Privacy Defenses

- **Robust aggregation**
 - Detects and rejects corrupted updates from a fraction of clients to protect against network instabilities and malicious actors.
- **Differential privacy**
 - Limits the information revealed about a single entry in a database by adding noise or using regularization techniques, either locally or centrally.
- **Homomorphic Encryption**
 - Encryption that allows computations to be performed on ciphertext without the need for decryption.
- **Privacy Preserving Federated Framework (PPFF)**
 - Comprehensive frameworks that implement known an original federated learning defenses to harden security.

Security Levels

Security Level	Integrity	Encryption	Untrusted Clients	Untrusted Server	Federated Defenses
Public	✓	X	X	X	X
Restricted	✓	✓	X	X	X
Sensitive	✓	✓	✓	X	Robust Aggregation
Secure	✓	✓	✓	✓	Robust Aggregation, Differential Privacy, Homomorphic Encryption, PPFF

Future Work

- Implementations of some or all federated defenses discussed in cross silo and cross device federated environments
- Measurements of time complexity within each security level to make sure lower security levels have lower time complexity
- Attempt known attacks on implementation to test for missing or outdated defenses



References

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- Mugunthan, Vaikkunth and Antigoni Polychroniadou. "SMPAI: Secure Multi-Party Computation for Federated Learning." (2019).
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THANKS!

Do you have any questions?

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